EE 533 Information Theory

Week 06

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Fixed Length Source Coding (Review)

A length n rate R source code $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$ uses nR bits to send n source symbol:

Encoder $\varphi^{(n)} : \mathcal{X}_1^n \to \{1, 2, \ldots, \lfloor 2^{nR} \rfloor\}$

Decoder $\widetilde{\varphi}^{(n)} : \{1, 2, \ldots, \lfloor 2^{nR} \rfloor\} \to \mathcal{X}_1^n$

Goal To minimize $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right)$ where $\mathcal{E}^{(n)} := \{x_1^n \in \mathcal{X}_1^n : x_1^n \neq \widetilde{\varphi}^{(n)}(\varphi^{(n)}(x_1^n))\}$

Motivation For i.i.d. $\mathsf{X}_1, \mathsf{X}_2, \mathsf{X}_3, \ldots$ determine the rate R for which there exists $(\varphi^{(n)}, \widetilde{\varphi}^{(n)})$'s s.t.

$$\lim_{n\to\infty} \mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) = 0 \qquad \text{for all } n \text{ large enough.}$$

Lemma (Achievability)

For any $R > H(\mathsf{X})$ and $\epsilon > 0$, $\exists n_0$ s.t. for all $n \geq n_0$ there exist a rate R code s.t. $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \leq \epsilon$.

Lemma (Converse)

For any rate R code $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \geq \left(H(\mathsf{X}) - R - \frac{1}{n}\right) \frac{1}{\log|\mathcal{X}|}$,

Lemma (Strong Converse)

For any $R < H(\mathsf{X})$ and $\epsilon > 0$, $\exists n_0$ s.t. for all $n \geq n_0$ for any rate R code s.t. $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \geq 1 - \epsilon$.

An Achievability Proof via Entropy Typicality

Lemma (Achievability)

For any $R > H(\mathsf{X})$ and $\epsilon > 0$, $\exists n_0$ s.t. for all $n \geq n_0$ there exist a rate R code s.t. $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \leq \epsilon$.

Proof.

$$\mathcal{A}_\gamma^{(n)} := \{x_1^n \in \mathcal{X}_1^n : 2^{-n(H(\mathsf{X})+\gamma)} \leq p_{\mathsf{X}_1^n}(x_1^n) \leq 2^{-n(H(\mathsf{X})-\gamma)}\}$$

Recall that $\forall \delta > 0$, $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\gamma^{(n)}\right) \geq 1 - \delta$ for sufficiently large n and set $\delta = \epsilon$.

$$\mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\gamma^{(n)}\right) \leq \epsilon \qquad \text{for sufficiently large } n \tag{1}$$

On the other hand $\left|\mathcal{A}_\gamma^{(n)}\right| \leq 2^{n(H(\mathsf{X})+\gamma)}$ and all elements of the typical set $\mathcal{A}_\gamma^{(n)}$ can be assigned to a distinct integer between 1 and $\lfloor 2^{nR} \rfloor$, for any γ less than $R - H(\mathsf{X})$.

$$\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)} \cap \mathcal{A}_\gamma^{(n)}\right) = 0 \qquad \text{for any } \gamma \in (0, R - H(\mathsf{X})) \text{ and } n \tag{2}$$

$$\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \leq \mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)} \cap \mathcal{A}_\gamma^{(n)}\right) + \mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\gamma^{(n)}(\mathsf{X})\right) \tag{3}$$

$\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \leq \epsilon$ for sufficiently large n by (1), (2), and (3). □

Strong converse proof used $p_{\mathsf{X}_1^n}(x_1^n) \leq 2^{-n(H(\mathsf{X})-\gamma)}$ for $x_1^n \in \mathcal{A}_\gamma^{(n)}$ instead of $\left|\mathcal{A}_\gamma^{(n)}\right| \leq 2^{n(H(\mathsf{X})+\gamma)}$

Remarks on the Achievability Proof via Entropy Typicality

- Will it make a difference if we map source strings, i.e. x_1^n's, with the highest probability, i.e. $p_{X_1^n}(x_1^n)$, to positive integers less than $\lceil 2^{nR} \rceil$ rather than the elements of the typical set?

  No in terms of the achievable rate: for any $\gamma > 0$

  $$H(X) + \gamma \geq \frac{1}{n} \log \left| \mathcal{B}_\epsilon^{(n)} \right| \geq H(X) - \gamma \qquad \text{for large enough n.}$$

  where $\mathcal{B}_\epsilon^{(n)} \subset \mathcal{X}_1^n$ is the smallest set, i.e. the one with the fewest number of elements, satisfying $\mathbf{P}\left(X_1^k \in \mathcal{B}_\epsilon^{(n)}\right) \geq 1 - \epsilon$.

  Yes in terms of the error probability at any finite value of n.
- How do we obtain sharper bounds on the number of bits necessary to reach a given error probability?

  Work with $\left| \mathcal{B}_\epsilon^{(n)} \right|$ instead of $\mathcal{A}_\gamma^{(n)}$ and invoke Berry-Esseen theorem instead of W.L.L.N.
- What is the smallest error probability one can achieve a $R > H(X)$?
- What is the smallest error probability one can achieve a $R < H(X)$?

The Error Probability at Rates Above The Source Entropy

Lemma

Let $X_1, X_2, \ldots$ be i.i.d. with p_X with a finite support, $R > H(X)$, and $\epsilon > 0$. For n large enough all rate R codes satisfy $P_e \geq e^{-n(E_{sp}(R,X)+\epsilon)}$ and there exists a rate R code satisfying $P_e \leq e^{-n(E_{sp}(R,X)-\epsilon)}$, where $E_{sp}(R,X) := \sup_{\alpha\in(0,1)} \frac{1-\alpha}{\alpha}(R - H_\alpha(X))$.

Proof.

$$\frac{d}{d\alpha} H_\alpha(X) = \frac{d}{d\alpha}\frac{1}{1-\alpha}\log\sum_x [p_X(x)]^\alpha$$

$$= -(1-\alpha)^{-2} D(p_{X_\alpha}\|p_X) \qquad \text{where } p_{X_\alpha}(x) = 2^{(\alpha-1)H_\alpha(X)}[p_X(x)]^\alpha$$

$$\frac{d}{d\alpha}\frac{1-\alpha}{\alpha}(R - H_\alpha(X)) = \frac{1}{\alpha^2}[H(X_\alpha) - R]$$

$$\frac{d}{d\alpha} H(X_\alpha) = \frac{-1}{\log e}\mathbf{E}\left[\left(\log\frac{1}{p_{X_\alpha}(X_\alpha)} - H(X_\alpha)\right)^2\right] \qquad \text{i.e., } H(X_\alpha) \text{ is decreasing in } \alpha$$

- If $R > \log|\mathcal{X}|$ then $E_{sp}(R,X) = \infty$ because $\lim_{\alpha\downarrow 0} H_\alpha(X) \uparrow \log|\mathcal{X}|$.
  $P_e \geq 0$ for all codes by definition. The exists a rate R code satisfying $P_e = 0$ by $|\mathcal{X}_1^n| \leq 2^{nR}$
- If $R < \log|\mathcal{X}|$, then $\exists\alpha^*$ satisfying $R = H(X_{\alpha^*})$ by the intermediate value theorem because $H(X_\alpha)$ is continuous in α, $\lim_{\alpha\downarrow 0} H(X_\alpha)\uparrow \log|\mathcal{X}|$, and $\lim_{\alpha\uparrow 1} H(X_\alpha)\downarrow H(X)$

$$E_{sp}(R,X) = D\left(p_{X_{\alpha^*}}\middle\|p_X\right)$$

Let $\tilde{\alpha} < \alpha^*$ be such that $D(p_{\mathsf{X}_{\tilde{\alpha}}} \| p_{\mathsf{X}}) \leq E_{sp}(R, \mathsf{X}) + \epsilon/3$ and $H(\mathsf{X}_{\tilde{\alpha}}) > R$, such an $\tilde{\alpha}$ exists because $H(\mathsf{X}_\alpha)$ is decreasing in α and $D(p_{\mathsf{X}_\alpha} \| p_{\mathsf{X}})$ is continuous in α.
Let $\mathsf{Z}_1, \mathsf{Z}_2, \ldots$ be i.i.d. with $p_{\mathsf{Z}} = p_{\mathsf{X}_{\tilde{\alpha}}}$.

$$\mathcal{D}_\gamma^{(n)} := \left\{ x_1^n \in \mathcal{X}_1^n : 2^{n(D(p_{\mathsf{X}_{\tilde{\alpha}}} \| p_{\mathsf{X}}) - \gamma)} \leq \frac{p_{\mathsf{Z}_1^n}(x_1^n)}{p_{\mathsf{X}_1^n}(x_1^n)} \leq 2^{n(D(p_{\mathsf{X}_{\tilde{\alpha}}} \| p_{\mathsf{X}}) + \gamma)} \right\}$$

$$\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \geq \mathbf{P}\left(\mathsf{Z}_1^n \in \mathcal{E}^{(n)} \cap \mathcal{D}_\gamma^{(n)}\right) 2^{-n(D(p_{\mathsf{X}_{\tilde{\alpha}}} \| p_{\mathsf{X}}) + \gamma)}$$

(i) $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \geq \mathbf{P}\left(\mathsf{Z}_1^n \in \mathcal{E}^{(n)} \cap \mathcal{D}_\gamma^{(n)}\right) 2^{-n(E_{sp}(R, \mathsf{X}) + 2\epsilon/3)}$ for $\gamma = \epsilon/3$

$\forall \delta > 0$, $\mathbf{P}\left(\mathsf{Z}_1^n \in \mathcal{D}_\gamma^{(n)}\right) \geq 1 - \delta$ for sufficiently large n by the W.L.L.N.

$\forall \delta > 0$, $\mathbf{P}\left(\mathsf{Z}_1^n \in \mathcal{E}^{(n)}\right) \geq 1 - \delta$ for sufficiently large n by the strong converse because $H(\mathsf{Z}) > R$.

(ii) $\forall \delta > 0$, $\mathbf{P}\left(\mathsf{Z}_1^n \in \mathcal{E}^{(n)} \cap \mathcal{D}_\gamma^{(n)}\right) \geq 1 - 2\delta$ for sufficiently large n.

$\forall \delta > 0$, $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \geq (1 - 2\delta) 2^{-n(E_{sp}(R, \mathsf{X}) + 2\epsilon/3)}$ for sufficiently large n by (i), (ii).

$\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \geq 2^{-n(E_{sp}(R, \mathsf{X}) + \epsilon)}$ for sufficiently large n.

Let $\hat{\alpha} > \alpha^*$ be such that $D(p_{X_{\hat{\alpha}}} \| p_X) \geq E_{sp}(R, X) - \epsilon$ and $H(X_{\hat{\alpha}}) < R$, such an $\hat{\alpha}$ exists because $H(X_\alpha)$ is decreasing in α and $D(p_{X_\alpha} \| p_X)$ is continuous in α.

$$\log \frac{1}{p_{X_\alpha}(x)} - H(X_\alpha) = \frac{\alpha}{1-\alpha}\left(\log \frac{p_{X_\alpha}(x)}{p_X(x)} - D(p_{X_\alpha} \| p_X)\right) \qquad \forall \alpha \in (0,1), \forall x \tag{4}$$

Let $Z_1, Z_2, \ldots$ be i.i.d. with $p_Z = p_{X_{\hat{\alpha}}}$.

$$\mathcal{F}_\gamma^{(n)} := \left\{ x_1^n \in \mathcal{X}_1^n : p_{Z_1^n}(x_1^n) \geq 2^{-n(H(X_{\hat{\alpha}})+\gamma)} \right\} \tag{5}$$

$$= \left\{ x_1^n \in \mathcal{X}_1^n : \frac{p_{Z_1^n}(x_1^n)}{p_{X_1^n}(x_1^n)} \leq 2^{n(D(p_{X_{\hat{\alpha}}} \| p_X) + \frac{1-\hat{\alpha}}{\hat{\alpha}}\gamma)} \right\} \qquad \text{by (4)} \tag{6}$$

$\left|\mathcal{F}_\gamma^{(n)}\right| \leq 2^{n(H(X_{\hat{\alpha}})+\gamma)}$ by (5) and $\mathbf{P}\left(Z_1^n \in \mathcal{F}_\gamma^{(n)}\right) \leq 1$.

Thus $\left|\mathcal{F}_\gamma^{(n)}\right| \leq 2^{nR - n\hat{\alpha}\epsilon}$ for $\gamma = R - H(X_{\hat{\alpha}}) - \hat{\alpha}\epsilon$

$\exists \varphi^{(n)}, \widetilde{\varphi}^{(n)}$ s.t. $\widetilde{\varphi}^{(n)}(\varphi^{(n)}(x_1^n)) = x_1^n$ for all $x_1^n \in \mathcal{F}_\gamma^{(n)}$, i.e. $\mathcal{E}^{(n)} \cap \left|\mathcal{F}_\gamma^{(n)}\right| = \emptyset$, for $\gamma = R - H(X_{\hat{\alpha}}) - \hat{\alpha}\epsilon$.

$$\begin{aligned}
\mathbf{P}\left(X_1^n \in \mathcal{E}^{(n)}\right) &= \mathbf{P}\left(X_1^n \notin \mathcal{F}_\gamma^{(n)}\right) \\
&\leq \mathbf{P}\left(Z_1^n \notin \mathcal{F}_\gamma^{(n)}\right) 2^{-n(D(p_{X_{\hat{\alpha}}} \| p_X) + \frac{1-\hat{\alpha}}{\hat{\alpha}}\gamma)} \qquad \text{by (6)} \\
&\leq 2^{-n(E_{sp}(R,X) - \hat{\alpha}\epsilon)}
\end{aligned}$$

The Error Probability at Rates Below The Source Entropy

Lemma

Let $X_1, X_2, \ldots$ be i.i.d. with p_X with a finite support, $R < H(X)$, and $\epsilon > 0$. For n large enough all rate R codes satisfy $P_e \geq 1 - e^{-n(E_{sc}(R,X)-\epsilon)}$ and there exists a rate R code satisfying $P_e \leq 1 - e^{-n(E_{sc}(R,X)+\epsilon)}$, where $E_{sc}(R,X) := \sup_{\alpha\in(1,\infty)} \frac{1-\alpha}{\alpha}(R - H_\alpha(X))$.

Proof.

$$\frac{d}{d\alpha} H_\alpha(X) = \frac{d}{d\alpha}\frac{1}{1-\alpha}\log\sum_x [p_X(x)]^\alpha$$

$$= -(1-\alpha)^{-2} D(p_{X_\alpha} \| p_X) \qquad \text{where } p_{X_\alpha}(x) = 2^{(\alpha-1)H_\alpha(X)}[p_X(x)]^\alpha$$

$$\frac{d}{d\alpha}\frac{1-\alpha}{\alpha}(R - H_\alpha(X)) = \frac{1}{\alpha^2}[H(X_\alpha) - R]$$

$$\frac{d}{d\alpha} H(X_\alpha) = \frac{-1}{\log e}\mathbf{E}\left[\left(\log\frac{1}{p_{X_\alpha}(X_\alpha)} - H(X_\alpha)\right)^2\right] \qquad \text{i.e., } H(X_\alpha) \text{ is decreasing in } \alpha$$

Let $\mathcal{X}_{\max} := \{x \in \mathcal{X} : p_X(x) = \max_{y\in\mathcal{X}} p_X(y)\}$

- If $R < \log|\mathcal{X}_{\max}|$ then $E_{sc}(R,X) = H_\infty(X) - R$ because $\lim_{\alpha\uparrow\infty} H_\alpha(X) \downarrow \log|\mathcal{X}_{\max}|$.
- If $R > \log|\mathcal{X}_{\max}|$, then $\exists\alpha^*$ satisfying $R = H(X_{\alpha^*})$ by the intermediate value theorem because $H(X_\alpha)$ is continuous in α, $\lim_{\alpha\downarrow 1} H(X_\alpha) \uparrow H(X)$, and $\lim_{\alpha\uparrow\infty} H(X_\alpha) \downarrow \log|\mathcal{X}_{\max}|$

$$E_{sc}(R,X) = D(p_{X_{\alpha^*}} \| p_X)$$

The Error Exponents For Fixed Length Source Entropy

Lemma

Let $X_1, X_2, \ldots$ *be i.i.d. with* p_X *with a finite support,* $R > H(X)$*, and* $\epsilon > 0$*. For* n *large enough all rate* R *codes satisfy* $P_e \geq e^{-n(E_{sp}(R,X)+\epsilon)}$ *and there exists a rate* R *code satisfying* $P_e \leq e^{-n(E_{sp}(R,X)-\epsilon)}$*, where* $E_{sp}(R,X) := \sup_{\alpha \in (0,1)} \frac{1-\alpha}{\alpha}(R - H_\alpha(X))$*.*

Lemma

Let $X_1, X_2, \ldots$ *be i.i.d. with* p_X *with a finite support,* $R < H(X)$*, and* $\epsilon > 0$*. For* n *large enough all rate* R *codes satisfy* $P_e \geq 1 - e^{-n(E_{sc}(R,X)-\epsilon)}$ *and there exists a rate* R *code satisfying* $P_e \leq 1 - e^{-n(E_{sc}(R,X)+\epsilon)}$*, where* $E_{sc}(R,X) := \sup_{\alpha \in (1,\infty)} \frac{1-\alpha}{\alpha}(R - H_\alpha(X))$*.*

Remark

Both for $R > H(X)$ *and* $R < H(X)$ *case results can be improved using and Berry-Esseen theorem or using sharper bounds [CL71].*

A Randomized Achievability Proof via Entropy Typicality

Lemma (Achievability)

For any $R > H(\mathsf{X})$ and $\epsilon > 0$, $\exists n_0$ s.t. for all $n \geq n_0$ there exist a rate R code s.t. $\mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{E}^{(n)}\right) \leq \epsilon$.

Proof.

Let us consider the following random choice of $(\varphi, \widetilde{\varphi})$

$\varphi(x_1^n)$ is assigned to an integer between 1 and $\lfloor 2^{nR} \rfloor$ uniformly at random independently of the assignments other elements of $\mathcal{X}_1^n$, i.e. $p_\Phi(\varphi) = \prod_{x_1^n \in \mathcal{X}_1^n} \frac{1}{\lfloor 2^{nR} \rfloor} = \left(1/\lfloor 2^{nR} \rfloor\right)^{|\mathcal{X}_1^n|}$.

Given a particular encoding function $\varphi : \mathcal{X}_1^n \to \{1, 2, \ldots, \lfloor 2^{nR} \rfloor\}$

$$\widetilde{\varphi}(k) = \begin{cases} x_1^n & \text{if } x_1^n \text{ is the only element of } \mathcal{A}_\gamma^{(n)} \text{ satifying } \varphi(x_1^n) = k \\ \text{error} & \text{else} \end{cases}$$

$$\mathcal{A}_\gamma^{(n)} := \{x_1^n \in \mathcal{X}_1^n : 2^{-n(H(\mathsf{X})+\gamma)} \leq p_{\mathsf{X}_1^n}(x_1^n) \leq 2^{-n(H(\mathsf{X})-\gamma)}\}$$

$$\mathbf{P}\left(\mathsf{X}_1^n \neq \widetilde{\Phi}(\Phi(\mathsf{X}_1^n))\right) \leq \mathbf{P}\left(\mathsf{X}_1^n \notin \mathcal{A}_\gamma^{(n)}\right) + \mathbf{P}\left(\mathsf{X}_1^n \in \mathcal{A}_\gamma^{(n)} \text{and } \exists z_1^n \in \mathcal{A}_\gamma^{(n)} \text{ s.t. } \Phi(\mathsf{X}_1^n) = \Phi(z_1^n)\right)$$

A Randomized Achievability Proof via Entropy Typicality cont'd

$$\mathbf{P}\left(X_1^n \neq \widetilde{\Phi}(\Phi(X_1^n))\right) \leq \mathbf{P}\left(X_1^n \notin \mathcal{A}_\gamma^{(n)}\right) + \mathbf{P}\left(X_1^n \in \mathcal{A}_\gamma^{(n)} \text{and } \exists z_1^n \in \mathcal{A}_\gamma^{(n)} \text{ s.t. } \Phi(X_1^n) = \Phi(z_1^n)\right)$$

Since $\forall \delta > 0$, $\mathbf{P}\left(X_1^n \in \mathcal{A}_\gamma^{(n)}\right) \geq 1 - \delta$ for sufficiently large n and set $\delta = \epsilon/2$.

$$\mathbf{P}\left(X_1^n \neq \widetilde{\Phi}(\Phi(X_1^n))\right) \leq \epsilon/2 + \mathbf{P}\left(X_1^n \in \mathcal{A}_\gamma^{(n)} \text{and } \exists z_1^n \in \mathcal{A}_\gamma^{(n)} \text{ s.t. } \Phi(X_1^n) = \Phi(z_1^n)\right)$$

Then by the union bound, i.e. by $\mathbf{P}(\cup_t \mathcal{B}_t) \leq \sum_t \mathbf{P}(\mathcal{B}_t)$,

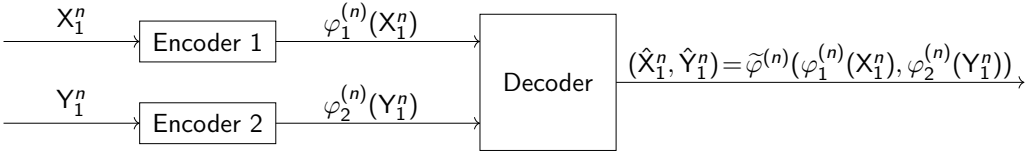
$$\mathbf{P}\left(X_1^n \neq \widetilde{\Phi}(\Phi(X_1^n))\right) \leq \epsilon/2 + \sum_{x_1^n \in \mathcal{A}_\gamma^{(n)}} \mathbf{P}(X_1^n = x_1^n) \sum_{z_1^n \in \mathcal{A}_\gamma^{(n)} \setminus \{x_1^n\}} \mathbf{P}(\Phi(X_1^n) = \Phi(z_1^n) | X_1^n = x_1^n)$$

$\mathbf{P}(\Phi(X_1^n) = \Phi(z_1^n) | X_1^n = x_1^n) = \mathbf{P}(\Phi(x_1^n) = \Phi(z_1^n))$ because Φ and X_1^n are independent.
$\mathbf{P}(\Phi(x_1^n) = \Phi(z_1^n)) = 1/\lfloor 2^{nR} \rfloor$ for any $z_1^n \neq x_1^n$ because $\Phi(z_1^n)$ and $\Phi(x_1^n)$ are i.i.d. on $\{1, \ldots, \lfloor 2^{nR} \rfloor\}$.

$$\begin{aligned}
\mathbf{P}\left(X_1^n \neq \widetilde{\Phi}(\Phi(X_1^n))\right) &\leq \epsilon/2 + 2\left|\mathcal{A}_\gamma^{(n)}\right| 2^{-nR} \\
&\leq \epsilon/2 + 2^{n(H(X)+\gamma)} 2^{1-nR} && \text{because } \left|\mathcal{A}_\gamma^{(n)}\right| \leq 2^{n(H(X)+\gamma)} \\
&\leq \epsilon && \text{for large enough } n \text{ if } \gamma < R - H(X)
\end{aligned}$$

Pairwise independence of $\Phi(z_1^n)$ and $\Phi(x_1^n)$ for $z_1^n \neq x_1^n$ suffices; joint independence is not needed.

Distributed Source Coding Problem



Let $(X_1, Y_1), (X_2, Y_2), \ldots$ be i.i.d. with $p_{X,Y}$. How many bits do we need per source symbol pair? What if X_i's and Y_i's are observed at physically separate locations?

$$\varphi_1^{(n)} : \mathcal{X}_1^n \to \{1, 2, \ldots, \lfloor 2^{nR_1} \rfloor\}$$
$$\varphi_2^{(n)} : \mathcal{Y}_1^n \to \{1, 2, \ldots, \lfloor 2^{nR_2} \rfloor\}$$
$$\widetilde{\varphi}^{(n)} : \{1, \ldots, \lfloor 2^{nR_1} \rfloor\} \times \{1, \ldots, \lfloor 2^{nR_2} \rfloor\} \to \mathcal{X}_1^n \times \mathcal{Y}_1^n$$

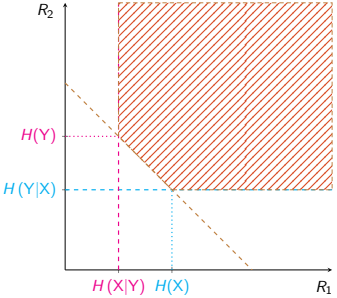
- The probability of error of a disributed source code $(\varphi_1^{(n)}, \varphi_2^{(n)}, \widetilde{\varphi}^{(n)})$ is
$$P_e^{(n)} := \mathbf{P}\left((X_1^n, Y_1^n) \neq \widetilde{\varphi}^{(n)}(\varphi_1^{(n)}(X_1^n), \varphi_2^{(n)}(Y_1^n))\right)$$
- A rate pair (R_1, R_2) is achievable iff $\exists (2^{nR_1}, 2^{nR_2}, n)$ distributed source codes with $P_e^{(n)} \to 0$.
- A achievable rate region is the closure of the set of achievable rates.

Slepian-Wolf Theorem for Distributed Source Coding

Theorem (Slepian-Wolf)

Let $(X_1, Y_1), (X_2, Y_2), \ldots$ *i.i.d. with* $p_{X,Y}$*, then for the corresponding distributed source coding problem the achievable rate region is given by*

$$R_1 \geq H(X|Y)$$
$$R_2 \geq H(Y|X)$$
$$R_1 + R_2 \geq H(X, Y)$$



Example ([CT06, Exampla 15.4.1])

$p(x, y)$	Metropolis: Rain	Metropolis: Shine
Gotham		
Rain	0.445	0.055
Shine	0.055	0.445

$$H(G) = 1 \qquad H(M) = 1$$
$$H(G|M) \approx 0.5 \qquad H(M|G) \approx 0.5$$

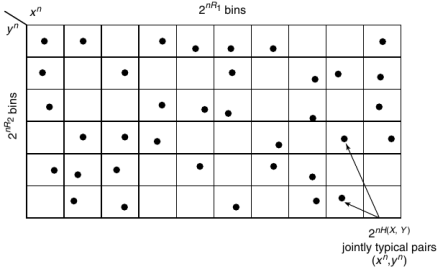
Achievability Proof for Slepian-Wolf Theorem - I

Let us consider the following random choice of $(\varphi_1, \varphi_2, \widetilde{\varphi})$

$\Phi_1(x_1^n)$'s are i.i.d. with uniform distribution on the integers between 1 and $\lfloor 2^{nR_1} \rfloor$ for $x_1^n \in \mathcal{X}_1^n$.

$\Phi_2(y_1^n)$'s are i.i.d. with uniform distribution on the integers between 1 and $\lfloor 2^{nR_2} \rfloor$ for $y_1^n \in \mathcal{Y}_1^n$.

$p_{\Phi_1,\Phi_2}(\varphi_1, \varphi_2) = \left(1/\lfloor 2^{nR_1} \rfloor\right)^{|\mathcal{X}_1^n|} \left(1/\lfloor 2^{nR_2} \rfloor\right)^{|\mathcal{Y}_1^n|}$.



Given a particular encoding functions $\varphi_1 : \mathcal{X}_1^n \to \{1, \ldots, \lfloor 2^{nR_1} \rfloor\}$ and $\varphi_2 : \mathcal{Y}_1^n \to \{1, \ldots, \lfloor 2^{nR_2} \rfloor\}$

$$\widetilde{\varphi}(k, l) = \begin{cases} (x_1^n, y_1^n) & \text{if } \exists!(x_1^n, y_1^n) \in \mathcal{A}_\gamma^{(n)} \text{ satifying both } \varphi_1(x_1^n) = k \text{ and } \varphi_2(y_1^n) = l \\ \text{error} & \text{else} \end{cases}$$

$$\mathcal{A}_\gamma^{(n)} := \left\{ (x_1^n, y_1^n) \in \mathcal{X}_1^n \times \mathcal{Y}_1^n \text{ satisfying } \left\{ \begin{array}{l} 2^{-n(H(X)+\gamma)} \leq p_{X_1^n}(x_1^n) \leq 2^{-n(H(X)-\gamma)}, \\ 2^{-n(H(Y)+\gamma)} \leq p_{Y_1^n}(y_1^n) \leq 2^{-n(H(Y)-\gamma)}, \text{and} \\ 2^{-n(H(X,Y)+\gamma)} \leq p_{X_1^n, Y_1^n}(x_1^n, y_1^n) \leq 2^{-n(H(X,Y)-\gamma)} \end{array} \right\} \right\}$$

Achievability Proof for Slepian-Wolf Theorem - II

$\mathbf{E}[P_e] \leq \mathbf{P}(\mathcal{E}_0) + \mathbf{P}(\mathcal{E}_1) + \mathbf{P}(\mathcal{E}_2) + \mathbf{P}(\mathcal{E}_3)$ by the union bound

$$\mathcal{E}_0 = \{(X_1^n, Y_1^n) \notin \mathcal{A}_\gamma^{(n)}\}$$

$$\mathcal{E}_1 = \{(X_1^n, Y_1^n) \in \mathcal{A}_\gamma^{(n)} \text{ and } \exists \tilde{x}_1^n \neq X_1^n \text{ s.t. } (\tilde{x}_1^n, Y_1^n) \in \mathcal{A}_\gamma^{(n)} \ \& \ \Phi_1(\tilde{x}_1^n) = \Phi_1(X_1^n)\}$$

$$\mathcal{E}_2 = \{(X_1^n, Y_1^n) \in \mathcal{A}_\gamma^{(n)} \text{ and } \exists \tilde{y}_1^n \neq Y_1^n \text{ s.t. } (X_1^n, \tilde{y}_1^n) \in \mathcal{A}_\gamma^{(n)} \ \& \ \Phi_2(\tilde{y}_1^n) = \Phi_2(Y_1^n)\}$$

$$\mathcal{E}_3 = \{(X_1^n, Y_1^n) \in \mathcal{A}_\gamma^{(n)} \text{ and } \exists (\tilde{x}_1^n, \tilde{y}_1^n) \in \mathcal{A}_\gamma^{(n)} : \tilde{x}_1^n \neq X_1^n, \Phi_1(\tilde{x}_1^n) = \Phi_1(X_1^n), \tilde{y}_1^n \neq Y_1^n, \Phi_2(\tilde{y}_1^n) = \Phi_2(Y_1^n)\}$$

$\mathbf{P}(\mathcal{E}_0) \leq \epsilon/4$ for sufficiently large n because $\forall \delta > 0$, $\mathbf{P}\left((X_1^n, Y_1^n) \in \mathcal{A}_\gamma^{(n)}\right) \geq 1 - \delta$ by A.E.P.

$$\mathbf{P}(\mathcal{E}_1) \leq \sum_{(x_1^n, y_1^n) \in \mathcal{A}_\gamma^{(n)}} p_{X_1^n, Y_1^n}(x_1^n, y_1^n) \sum_{\tilde{x}_1^n \in \mathcal{A}_\gamma^{(n)}(X|y_1^n) \setminus \{x_1^n\}} \mathbf{P}(\Phi_1(x_1^n) = \Phi_1(\tilde{x}_1^n) | X_1^n = x_1^n, Y_1^n = y_1^n)$$

$$\leq \sum_{(x_1^n, y_1^n) \in \mathcal{A}_\gamma^{(n)}} p_{X_1^n, Y_1^n}(x_1^n, y_1^n) \left|\mathcal{A}_\gamma^{(n)}(X|y_1^n)\right| 2^{-nR_1}$$

where $\mathcal{A}_\gamma^{(n)}(X|y_1^n) := \{x_1^n : (x_1^n, y_1^n) \in \mathcal{A}_\gamma^{(n)}(X, Y)\}$. Then $\left|\mathcal{A}_\gamma^{(n)}(X|y_1^n)\right| \leq 2^{n(H(X|Y)+2\gamma)}$ because $p_{X_1^n|Y_1^n}(\tilde{x}_1^n|y_1^n) \geq 2^{-n(H(X|Y)+2\gamma)}$ by $p_{X_1^n, Y_1^n}(\tilde{x}_1^n, y_1^n) \geq 2^{-n(H(X,Y)+\gamma)}$ and $p_{Y_1^n}(y_1^n) \leq 2^{-n(H(Y)-\gamma)}$.

$\mathbf{P}(\mathcal{E}_1) \leq \epsilon/4$ for large enough n provided that $\gamma < \frac{R_1 - H(X|Y)}{2}$

Achievability Proof for Slepian-Wolf Theorem - III

$$\mathbf{P}(\mathcal{E}_2) \leq \sum_{(x_1^n,y_1^n)\in\mathcal{A}_\gamma^{(n)}} p_{X_1^n,Y_1^n}(x_1^n,y_1^n) \sum_{\tilde{y}_1^n\in\mathcal{A}_\gamma^{(n)}(Y|x_1^n)\setminus\{y_1^n\}} \mathbf{P}(\Phi_2(y_1^n)=\Phi_2(\tilde{y}_1^n)|X_1^n=x_1^n, Y_1^n=y_1^n)$$

$$\leq \sum_{(x_1^n,y_1^n)\in\mathcal{A}_\gamma^{(n)}} p_{X_1^n,Y_1^n}(x_1^n,y_1^n) \left|\mathcal{A}_\gamma^{(n)}(Y|x_1^n)\right| 2^{-nR_2}$$

where $\mathcal{A}_\gamma^{(n)}(Y|x_1^n) := \{y_1^n : (x_1^n,y_1^n)\in\mathcal{A}_\gamma^{(n)}(X,Y)\}$. Then $\left|\mathcal{A}_\gamma^{(n)}(Y|x_1^n)\right| \leq 2^{n(H(Y|X)+2\gamma)}$ because $p_{Y_1^n|X_1^n}(\tilde{y}_1^n|x_1^n) \geq 2^{-n(H(Y|X)+2\gamma)}$ by $p_{X_1^n,Y_1^n}(x_1^n,\tilde{y}_1^n) \geq 2^{-n(H(X,Y)-\gamma)}$ and $p_{X_1^n}(x_1^n) \leq 2^{-n(H(X)+\gamma)}$.

$\mathbf{P}(\mathcal{E}_2) \leq \epsilon/4$ for large enough n provided that $\gamma < \frac{R_2-H(Y|X)}{2}$

$$\mathbf{P}(\mathcal{E}_3) \leq \sum_{(x_1^n,y_1^n)\in\mathcal{A}_\gamma^{(n)}} p_{X_1^n,Y_1^n}(x_1^n,y_1^n) \sum_{(\tilde{x}_1^n,\tilde{y}_1^n)\in\mathcal{A}_\gamma^{(n)}:\tilde{x}_1^n\neq x_1^n,\tilde{y}_1^n\neq y_1^n} \mathbf{P}(\Phi_1(x_1^n)=\Phi_1(\tilde{x}_1^n),\Phi_2(y_1^n)=\Phi_2(\tilde{y}_1^n)|X_1^n=x_1^n,Y_1^n=y_1^n)$$

$$\leq \left|\mathcal{A}_\gamma^{(n)}\right| 2^{-nR_1} 2^{-nR_2}$$

On the other hand $\left|\mathcal{A}_\gamma^{(n)}\right| \leq 2^{n(H(X,Y)+\gamma)}$ because $p_{X_1^n,Y_1^n}(\tilde{x}_1^n,\tilde{y}_1^n) \geq 2^{-n(H(X,Y)+\gamma)}$.

$\mathbf{P}(\mathcal{E}_3) \leq \epsilon/4$ for large enough n provided that $\gamma < R_1 + R_2 - H(X,Y)$

$\mathbf{E}[P_e] \leq \epsilon$ for large enough n for $\gamma = \min\{\frac{R_1-H(X|Y)}{3}, \frac{R_2-H(Y|X)}{3}, \frac{R_1+R_2-H(X,Y)}{3}\}$

Proof of the Converse for Slepian-Wolf Theorem - I

According to Slepian-Wolf theorem asserts that a rate pair (R_1, R_2) is outside the achievable rate region iff at least one of the following three inequalities hold.

$$R_1 < H(\mathsf{X}|\mathsf{Y}) \tag{*}$$

$$R_2 < H(\mathsf{Y}|\mathsf{X}) \tag{**}$$

$$R_1 + R_2 < H(\mathsf{X}, \mathsf{Y}) \tag{***}$$

Fano's inequality: $H(\mathsf{Z}|\mathsf{S}) \leq \hbar(P_{\mathbf{e}}) + P_{\mathbf{e}} \log |\mathcal{Z}|$ where $P_{\mathbf{e}} = \mathbf{P}(\mathsf{Z} \neq \mathsf{S})$ and Z, S are $\mathcal{Z}$-valued.

$$\begin{aligned}
nH(\mathsf{X},\mathsf{Y}) &= H(\mathsf{X}_1^n, \mathsf{Y}_1^n) && \text{by the joint entropy of ind. r.v.'s} \\
&= H(\mathsf{X}_1^n, \mathsf{Y}_1^n, \varphi_1(\mathsf{X}_1^n), \varphi_2(\mathsf{Y}_1^n)) && \text{chain rule and } H(g(\mathsf{Z})|\mathsf{Z}) = 0 \\
&= H(\varphi_1(\mathsf{X}_1^n), \varphi_2(\mathsf{Y}_1^n)) + H(\mathsf{X}_1^n, \mathsf{Y}_1^n | \varphi_1(\mathsf{X}_1^n), \varphi_2(\mathsf{Y}_1^n)) && \text{chain rule} \\
&\leq nR_1 + nR_2 + H(\mathsf{X}_1^n, \mathsf{Y}_1^n | \varphi_1(\mathsf{X}_1^n), \varphi_2(\mathsf{Y}_1^n)) && \text{bc. the range of } \varphi_i \text{ is } \{1, \ldots, 2^{nR_i}\}. \\
&\leq nR_1 + nR_2 + H(\mathsf{X}_1^n, \mathsf{Y}_1^n | \widetilde{\varphi}(\varphi_1(\mathsf{X}_1^n), \varphi_2(\mathsf{Y}_1^n))) && \text{conditioning reduces entropy} \\
&\leq n(R_1 + R_2) + \hbar(P_{\mathbf{e}}) + nP_{\mathbf{e}} \log |\mathcal{X} \times \mathcal{Y}| && \text{by Fano's inequality}
\end{aligned}$$

If (***) holds, ${P_{\mathbf{e}}}^{(n)}$ is bounded away from zero for large enough n.

Proof of the Converse for Slepian-Wolf Theorem - II

$$
\begin{aligned}
nH(\mathrm{X}|\mathrm{Y}) &= H(\mathrm{X}_1^n|\mathrm{Y}_1^n) \\
&= H(\mathrm{X}_1^n, \varphi_1(\mathrm{X}_1^n)|\mathrm{Y}_1^n) && \text{chain rule and } H(g(\mathrm{Z})|\mathrm{Z}) = 0 \\
&= H(\varphi_1(\mathrm{X}_1^n)|\mathrm{Y}_1^n) + H(\mathrm{X}_1^n|\mathrm{Y}_1^n, \varphi_1(\mathrm{X}_1^n)) && \text{chain rule} \\
&\leq nR_1 + H(\mathrm{X}_1^n|\mathrm{Y}_1^n, \varphi_1(\mathrm{X}_1^n)) && \text{by } \varphi_1(\mathrm{X}_1^n) \in \{1, \ldots, 2^{nR_1}\} \\
&\leq nR_1 + H(\mathrm{X}_1^n|\widetilde{\varphi}(\varphi_1(\mathrm{X}_1^n), \varphi_2(\mathrm{Y}_1^n))) && \text{conditioning reduces entropy} \\
&\leq nR_1 + \hbar\left(\mathbf{P}\left(\hat{\mathrm{X}}_1^n \neq \mathrm{X}_1^n\right)\right) + n\mathbf{P}\left(\hat{\mathrm{X}}_1^n \neq \mathrm{X}_1^n\right)\log|\mathcal{X}| && \text{by Fano's inequality} \\
&\leq nR_1 + 1 + nP_{\mathbf{e}} \log|\mathcal{X}| && \text{by } \hbar(z) \leq 1,\ \mathbf{P}\left(\hat{\mathrm{X}}_1^n \neq \mathrm{X}_1^n\right) \leq P_{\mathbf{e}}
\end{aligned}
$$

If (*) holds, ${P_{\mathbf{e}}}^{(n)}$ is bounded away from zero for large enough n.
Following a similar analysis one can show that if (**) holds, ${P_{\mathbf{e}}}^{(n)}$ is bounded away from zero for large enough n.
Thus a rate pair satifying any of (*), (**), (***) has to be outside the achievable rate region as claimed by the Slepian-Wolf theorem.

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